

Ad Safety Moderation Pipeline

Strict two-page technical synopsis | MS_DSP_462 Computer Vision

Measured results | proposal benchmarks labeled

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1. PROBLEM AND OBJECTIVES

Static ad creatives can contain restricted visual objects, visible promotion language, or ambiguous context. This pilot converts one local image into an auditable **APPROVE**, **REVIEW**, or **BLOCK** triage result. It does not infer legality, audience eligibility, landing-page claims, or jurisdictional compliance.

- Classify four bounded visual labels on an untouched grouped test split.
- Fuse optional object and OCR evidence without allowing auxiliary cues to trigger an automatic block.
- Expose the verdict, raw and fused score distributions, evidence provenance, stage timing, overlays, and a downloadable audit record.

2. DATASET AND PREPROCESSING

The formal registry contains **288** images in **215** source or campaign groups. Grouped splitting produced train/validation/test counts of 192/48/48. Verified overlap across splits: 0 source groups and 0 exact hashes.

Label	Train	Val	Test	Total
Safe / general advertising	48	12	12	72
Firearms and ammunition	48	12	12	72
Explosives and explosive devices	48	12	12	72
Financial or cryptocurrency promotion	48	12	12	72

Registry-observed dimensions span 253-768 px wide and 111-768 px high. Encodings: image/jpeg: 288. Synthetic-declared rows: 144/288. Sources: rajshivanshuu/weapons_set1 (144), EthanGabis/ADautoGen-DS (72), project_generated_financial_creatives_v1 (72).

- The formal registry audit confirms 288 hash-verified JPEGs with a maximum saved side of 768 pixels.
- Source or generated campaign groups stay within one split; the exact-file and group overlap counts are both zero.
- Decoded RGB images use the pinned timm backbone transform before frozen feature extraction; scaling is learned on train embeddings only.

3. MODEL ARCHITECTURE AND MATH

A pinned ViT-B/16 backbone (85,798,656 parameters, all frozen) converts each RGB creative into a 768-dimensional embedding. A StandardScaler and multinomial logistic head produce four class probabilities.

Grounding DINO and Tesseract are optional evidence channels. The policy layer applies noisy-OR fusion to restricted cues; only the calibrated classifier can trigger an automatic firearms or explosives block.

$$z = f_{\theta}(x) \text{ in } \mathbb{R}^{768}; u = (z - \mu) / \sigma$$
$$p(y=k|x) = \exp(w_k^T u + b_k) / \sum_j \exp(w_j^T u + b_j)$$
$$\text{fused_score}_k = 1 - \text{product}_i (1 - \text{evidence}_{k,i})$$

4. TRAINING AND HYPERPARAMETERS

Selected runtime model: **Frozen ViT + logistic head** (vit-frozen-063c6c38a5d8-logreg-v1). Test data remained outside fitting and threshold selection.

Split isolation	192 train fit; 48 validation calibration; 48 test once
Feature run	768-D; batch 8; cpu
Head	StandardScaler + multinomial logistic regression
Head settings	C=0.1; lbfgs; balanced; max_iter=5000
Frozen backbone	85.80M parameters; 0 trainable
Artifact thresholds	firearms 0.9914; explosives 0.1728; financial promotion 0.9946
Threshold tuning	validation recall >=0.90; distinct from proposal acceptance
Recorded timing	features 8.298s; head fit 0.0086s
Policy	1.0.0-pilot
Auxiliary review	fused restricted evidence 0.35; margin 0.12

5. DETAILED MEASURED PERFORMANCE

Untouched grouped test: n=48 (12 per class). Macro AP=1.000; group-bootstrap macro F1 95% interval=0.937 to 1.000 (2,000 source-group replicates).

TEST ACCURACY	MACRO F1
97.9%	97.9%
47 / 48 correct	four classes
ARGMAX RESTRICTED MEAN	THRESHOLD-OP MEAN
0.972222	0.944444
multiclass class-average	validation-selected thresholds
SAFE-CLASS PRECISION	SAFE-AD FALSE FLAG
1.000	0.0%
proposal >0.98: passes	1 - Safe recall

The 0.972222 figure is the multiclass-argmax class-average recall across firearms, explosives, and financial promotion; it is not the final threshold operating point. Safe-class precision is 1.000 and passes the proposal >0.98 precision target; it is distinct from the 0.0% Safe-ad false-flag rate (1 - Safe recall). Threshold selection used a >=0.90 validation-recall tuning floor, distinct from the proposal per-class >0.95 acceptance target. At thresholds selected on validation, untouched-test recalls are firearms 0.916667, explosives 0.916667, and financial promotion 1.000000; mean 0.944444, worst class 0.916667. The proposal criterion of per-class recall >0.95 fails for firearms and explosives.

Class	P	R	F1	n
Safe	1.000	1.000	1.000	12
Firearms	0.923	1.000	0.960	12
Explosives	1.000	0.917	0.957	12
Financial promo	1.000	1.000	1.000	12

Untouched test confusion matrix (counts)

True / pred	Safe	Fire	Expl	Fin
Safe	12	0	0	0
Firearms	0	12	0	0
Explosives	0	1	11	0
Financial promo	0	0	0	12

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6. BASELINE AND BACKBONE COMPARISON

The comparison below is reproduced from the saved model-selection artifact. Its column labels preserve whether each score is validation or test evidence.

Model	Val F1	Test F1	Test acc	CPU p95
ViT primary	0.958	0.979	0.979	71.5 ms
ResNet50 baseline	0.957	0.873	0.875	40.1 ms

7. MEASURED LATENCY

MEDIAN	CLASSIFIER-PATH P95
69.1 ms	71.527 ms
batch 1	batch 1; excludes auxiliaries
MEAN	SERIAL THROUGHPUT
68.7 ms	14.6 img/s
single thread	derived; not concurrent

The measured ViT classifier-path p95 is 71.527 ms and exceeds the proposal's <50 ms per-asset benchmark. Measured on Apple M2 Max / cpu: 3 warmups, 20 runs. Full end-to-end p95 was not measured because file decoding, OCR, object detection, network transfer, application rendering are excluded. Concurrent throughput and MPS resource use were not measured. Artifact scope: decoded in-memory PIL image; preprocessing + frozen backbone + logistic head.

8. CHALLENGES AND IMPLEMENTED SOLUTIONS

Source-style confounding remains.

The implementation grouped campaigns before splitting and recomputed all 288 hashes; this controls direct leakage but cannot remove source cues.

Threshold leakage could inflate safety claims.

The head fit on train only, restricted thresholds were selected on validation only, and the 48-image test was scored once afterward.

One grenade crossed the visual taxonomy boundary.

The saved failure was explosives classified as firearms (1/48 test images). The failure remains in the audit table; ambiguous restricted evidence routes to review rather than being hidden.

The measured classifier path misses the proposal latency benchmark.

ViT classifier-path CPU p95 was 71.527 ms, above 50 ms, versus 40.1 ms for the CNN baseline. Neither value is a full end-to-end p95.

EXTERNAL DIAGNOSTIC ONLY

manually reviewed external Wikimedia diagnostic spot check; not a formal test set Detector image recall was 100.0%, but the non-weapon detector false-positive rate was 90.0%; this is a generalization warning, not formal test evidence.

SAMPLES	CLASSIFIER ACCURACY
26	65.4%
MACRO F1	DETECTOR NONWEAPON FP
55.5%	90.0%

9. LIMITATIONS AND DEPLOYMENT BOUNDARY

- Every policy class is tied to one source or generator, so source-style confounding can inflate metrics.
- The untouched test set has only 12 images per class and cannot establish production performance.
- ADautoGen safe images and all financial creatives are synthetic, while weapons_set1 is not declared synthetic.
- weapons_set1 lacks per-image provenance and rights; repository-level OSL-3.0 metadata is not enough for production reuse.
- The weapons images are object-centric, while real ads may contain tiny, occluded, stylized, or contextual objects.
- The four-way label is single-class and omits mixed-policy ads, alcohol, tobacco, nudity, violence, and other proposal categories.
- A classifier score is visual evidence, not proof that an ad violates law or platform policy.
- Latency excludes file decoding, OCR, object detection, network transfer, and application rendering.
- Attention and occlusion maps are qualitative evidence, not causal explanations.

10. REPRODUCIBILITY AND EVIDENCE

Run	2026-08-25T18:28:58.613303+00:00 seed 462
Host	macOS-26.5.2-arm64-arm-64bit feature device cpu
Registry SHA-256	c5bad7b8d8ccbb8b...
Backbone revision	063c6c38a5d8510b...
Artifact SHA-256	3473f8f6817a007f...
Independent check	288 image hashes; reload max diff 0

Every reported metric comes from outputs/evaluation. The formal dataset claim comes only from data/capstone_registry.csv. The rejected Wikimedia registry is excluded from training and test evidence.

The Streamlit workspace visualizes the current case and saved benchmark context; its session charts are descriptive interface evidence, not production throughput, monitoring, or population-level performance evidence.

11. REFERENCES

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